FRONTEND DEVELOPER ASSESSMENT

Fomi — AI Content Generation Workspace

Part A implements the mockup as a responsive, production-quality workspace. Part B, **Fomi Studio**, redesigns Image Generation as version control: branch, diff and compare every result.

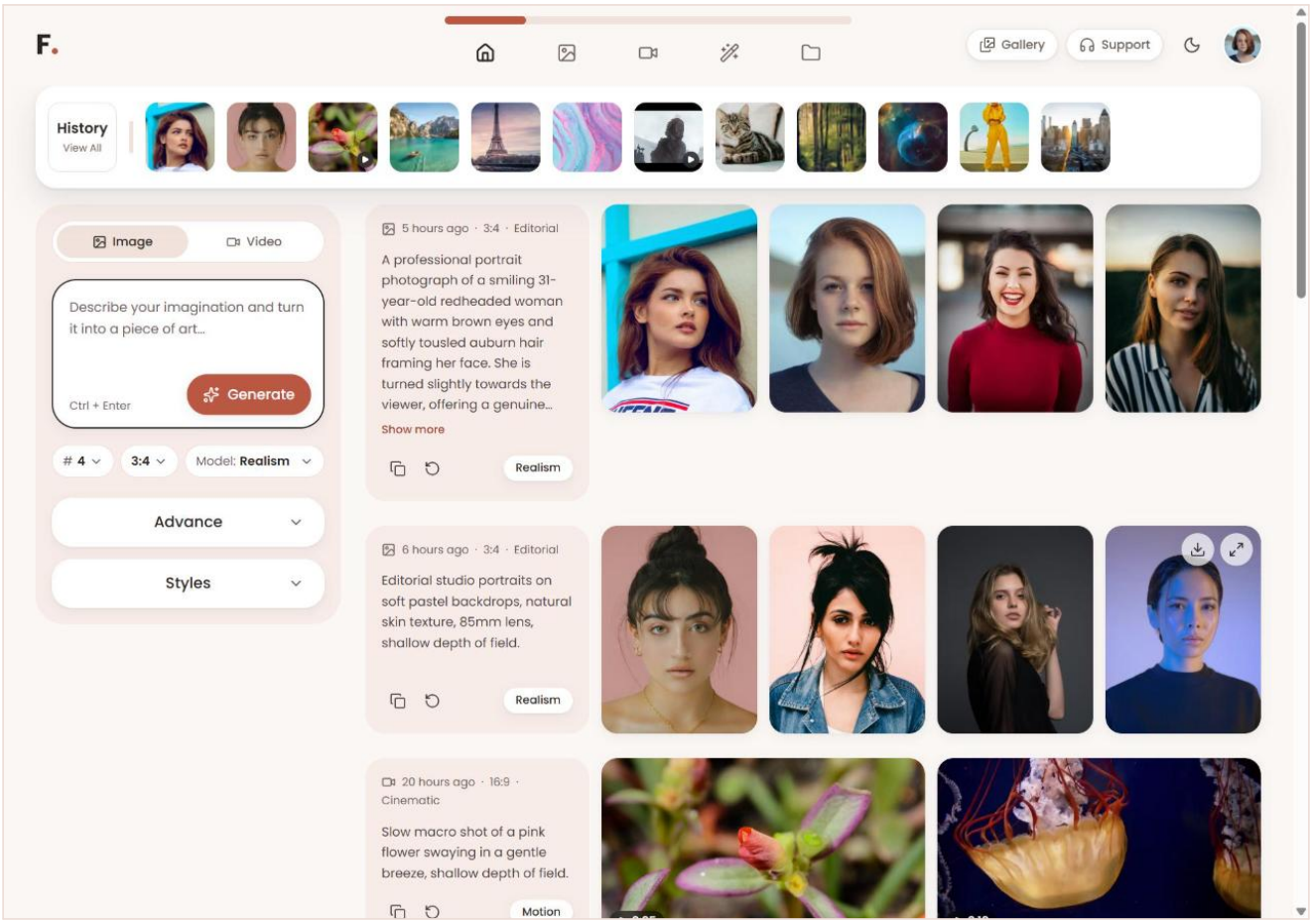
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Live — Part A workspace	ai-content-generation-xi.vercel.app
Live — Part B Studio	ai-content-generation-xi.vercel.app/studio
Source code (GitHub)	github.com/AwaisAmin/ai-content-generation
README (architecture, a11y, API)	github.com/.../ai-content-generation#readme

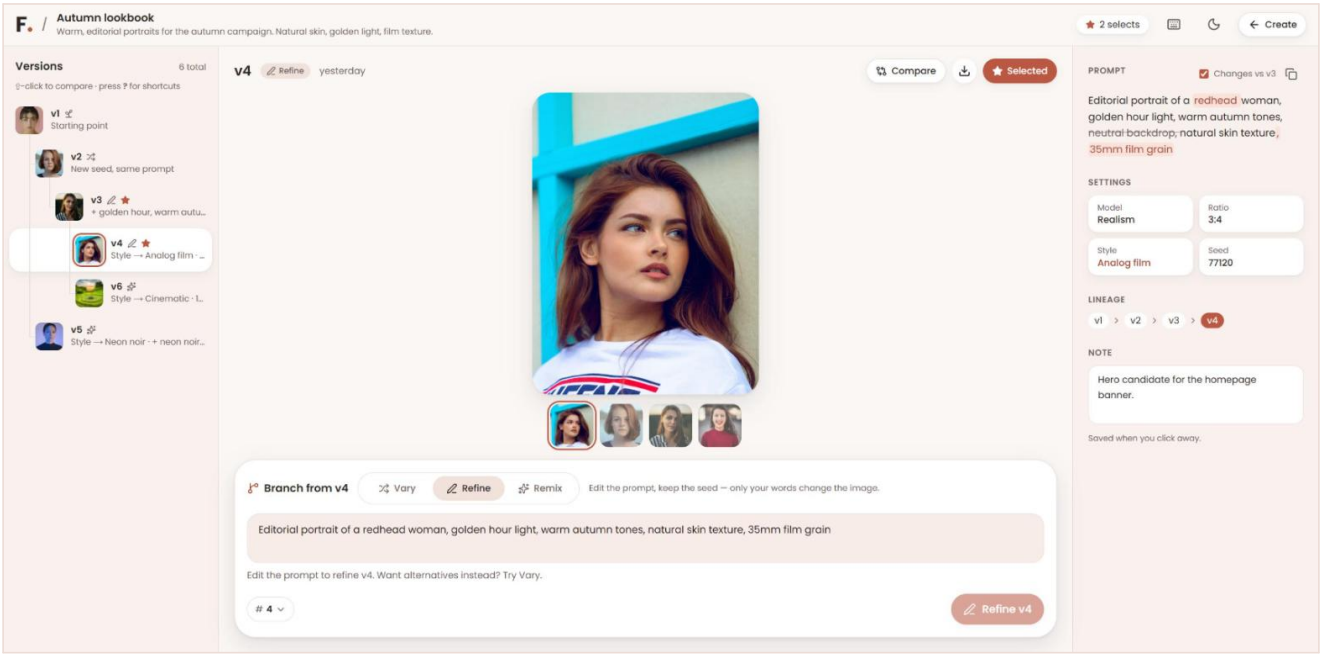
How to review in two minutes

- **Workspace** (`/`): type a prompt and press **Generate** or **Ctrl + Enter**. Add `#error` to see the failure and retry flow. Click any history thumbnail to jump to it, use **Reuse** on a prompt card, then try phone width and dark mode.
- **Studio** (`/studio`): select **v4**, press **R**, add a few words and watch the live diff, then **Ctrl + Enter**. The new version keeps v4's seed. Press **C** to compare with the parent and switch to **Slider**. Press **?** for all shortcuts.
- Built with **Next.js 16 (App Router)**, JavaScript, Tailwind CSS v4 and CSS Modules, with a mocked Route Handler API. WCAG AA contrast and keyboard support, audited with axe-core.

Screens



Part A — the generation workspace from the mockup, with history strip, prompt panel and results feed.



Part B — Fomi Studio: version tree, stage with hero output, branch composer with live diff, and inspector.

Fomi Studio — Product Thinking

Workspace redesigned: Image Generation
Live: ai-content-generation-xi.vercel.app/studio

Fomi Studio treats image generation like version control. Every result is a version with a parent, the operation that produced it, and a visible diff. You explore by **branching**, not by scrolling.

1. What problem were you trying to solve?

Professional creators don't generate once. They iterate dozens of times towards a brief, usually with a client or art director in the loop. Today's generators show that work as one long chronological feed, so:

- **Causality is lost.** Which change in the prompt made the light better? The feed can't tell you, because every result is just "newer".
- **Going back is costly.** Returning to a promising direction from yesterday means scrolling, then copying and pasting the old prompt.
- **Results feel random.** Changing words also changes the seed, so you can't tell if the improvement came from your edit or from luck.
- **Decisions aren't recorded.** Why was a version chosen, and what did the client say? That lives in someone's head or in Slack.

Who it's for: a professional — designer, art director, agency creative — who spends hours a day in the tool, returns to ongoing projects, and has to explain their choices to someone else.

2. What are the three most important UX decisions you made?

1. Versions form a tree, not a feed. Each version records its parent and how it was made, and the tree in the sidebar shows the whole exploration at a glance, with one-line summaries such as "+ golden hour, warm autumn tones". Returning users see exactly where they left off, which branches they abandoned, and which they starred, with a note explaining why. *Why:* a creative process naturally branches, and a linear feed throws that structure away.

2. Three explicit operations with predictable rules.

- **Vary** — same prompt and settings, new seed: explore alternatives.
- **Refine** — edit the prompt, **keep the seed**: only your words change the image.
- **Remix** — change style, model or ratio with a new seed.

Why: separating "try again" from "change what I asked for" makes iterating deliberate instead of a slot machine. Keeping the seed on Refine is the key detail: it isolates the effect of the words you changed.

3. Show the diff everywhere.

- The composer shows a **live diff before you generate**, so you see what you are about to change before spending a credit.
- The tree summarises each version's change in one line.
- The inspector highlights the prompt diff and any changed settings.
- **Compare** puts two versions side by side, or under a reveal slider, with the prompt and settings differences listed below.

Why: seeing cause next to effect is how professionals build intuition about prompts.

Supporting these: a **keyboard-first flow** (J/K to move, V/R/X to branch, C to compare, S to select, ? for help), and drawers on tablet and mobile so the same model works on any screen.

3. Which features did you intentionally leave out?

Left out	Why
A community / inspiration feed in the workspace	It competes for attention with the user's own work. Inspiration has its place (Gallery); the workspace is for producing.
A one-click "enhance my prompt" button	It rewrites the prompt invisibly, which breaks the core promise that you can see exactly what changed. It could return later as a suggestion shown as a diff.
A free-form node canvas	Powerful, but heavy to learn and maintain. A tree covers most real iteration with far less complexity.
A chat interface	Conversation loses structure: you can't branch from message 7 and compare it with message 12.
Collaboration, comments and sharing	Important for agencies, but it multiplies scope. Notes and selects cover the solo workflow first.
Credits, billing and model marketplaces	Not part of the core creative loop being tested here.

4. Which existing products inspired your thinking?

Product	What I liked	What I'd improve
Git / GitHub	Commits, branches and diffs make history understandable.	Too abstract for visual work, so I kept the ideas (parent, operation, diff) and dropped the jargon.
Cursor	The inline composer, diffs before applying changes, and a keyboard-first flow.	Its diffs are text-only. For images, the words need to sit next to the pictures they produced, which is what Compare does.
Figma	Version history and a calm, panel-based workspace.	Its history is linear and built for recovery rather than exploration.
Lightroom	Picks and stars, compare/survey views, and virtual copies.	Copies don't record *why* they differ.
Midjourney / Higgsfield	Fast Vary/Upscale actions on every result.	Lineage disappears into the feed after a few rounds.
Krea	Real-time feedback while you type.	No structure for the history behind a result.

5. If you had another month, what would you improve?

1. **Real model semantics:** true seed-locked Refine and image-to-image refinement of a masked region ("refine just the hair").
2. **Merging branches:** combine the lighting of v3 with the styling of v5, shown as a three-way diff.
3. **Scale:** collapsible subtrees, search and a minimap for projects with hundreds of versions, plus a virtualised tree.

4. **Review workflow:** share a read-only branch link, let clients comment on a version, and export selects to Assets.
5. **Persistence and safety:** a real database, undo, and archiving abandoned branches.
6. **Quality:** unit tests for the diff and tree logic, Playwright end-to-end tests for branching and compare, and usability sessions with working designers.

6. What single feature or workflow differentiates your workspace?

Prompt lineage with diffs. You can branch from any version with a predictable operation and see exactly what changed: live before you generate, and side by side afterwards. Other tools show you results. Fomi Studio shows you **why** a result is better, which turns prompting into a craft you can learn, repeat and explain to a client.

Trade-offs

- **Prioritised:** the core loop of select → branch → compare → select, and making it fast from the keyboard.
- **Simplified:** an indented tree instead of a free-form graph; one project; a mocked backend that picks images by prompt keywords.
- **Removed:** chat, a canvas, an inspiration feed and prompt auto-enhance. Each one added surface area without strengthening the core idea.